A HACKATHON PRIVACY PROTOTYPE

Veil

The agent works.
The tool never sees the originals.

Middleware between an AI agent and its tool. Detect. Tokenize. Verify. Restore. Then prove it with the exact payload the tool received.

TICKET-DEMO-1042

synthetic data only // deterministic offline // optional gemini

CONTENTS - NINE PAGES, ONE STORY

Start anywhere.

01 **The problem**
Agents leak what they carry

02 **Three moves**
Tokenize. Verify. Restore.

03 **Architecture**
The gateway is the only path

04 **Live demo**
Ninety seconds, zero leaks

05 **Proof**
Receipts, not promises

06 **Limits + run it**
Honest scope, two terminals

Agents leak *what they carry.*

Every helpful agent forwards your data verbatim - to tools, logs and vendors that never needed it.

YOU PASTE IT

An address, a phone number, an ID in a support chat.

IT FANS OUT

Tool payloads, logs and echoe keep the originals forever.

JUDGES ASK

"Show me what the tool actually received."

Tokenize. Verify. Restore.

01

TOKENIZE

Recursive text + JSON scan. 11 patterns become tokens like [[EMAIL:V..]].

02

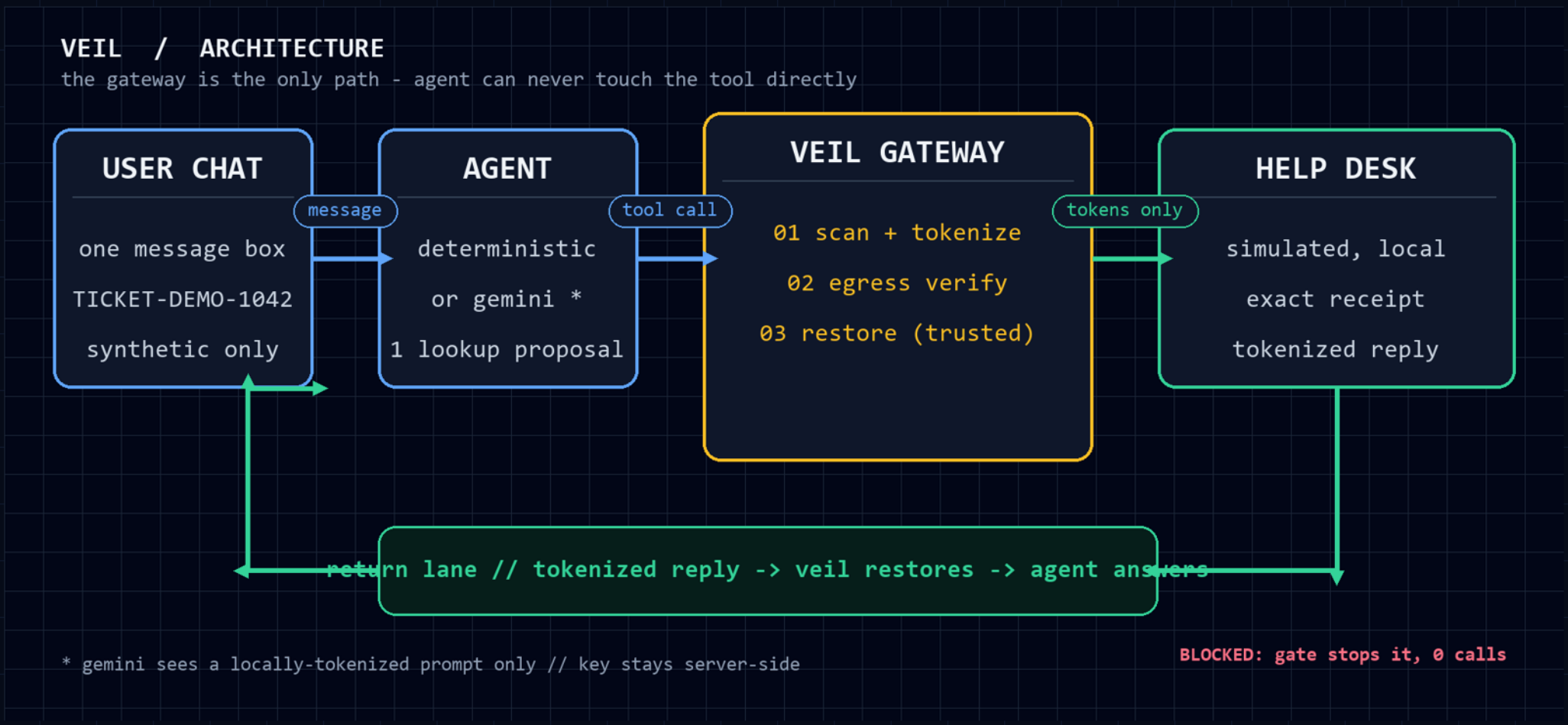
VERIFY

Egress gate re-scans the exact payload. Fail-closed: residue = zero calls.

03

RESTORE

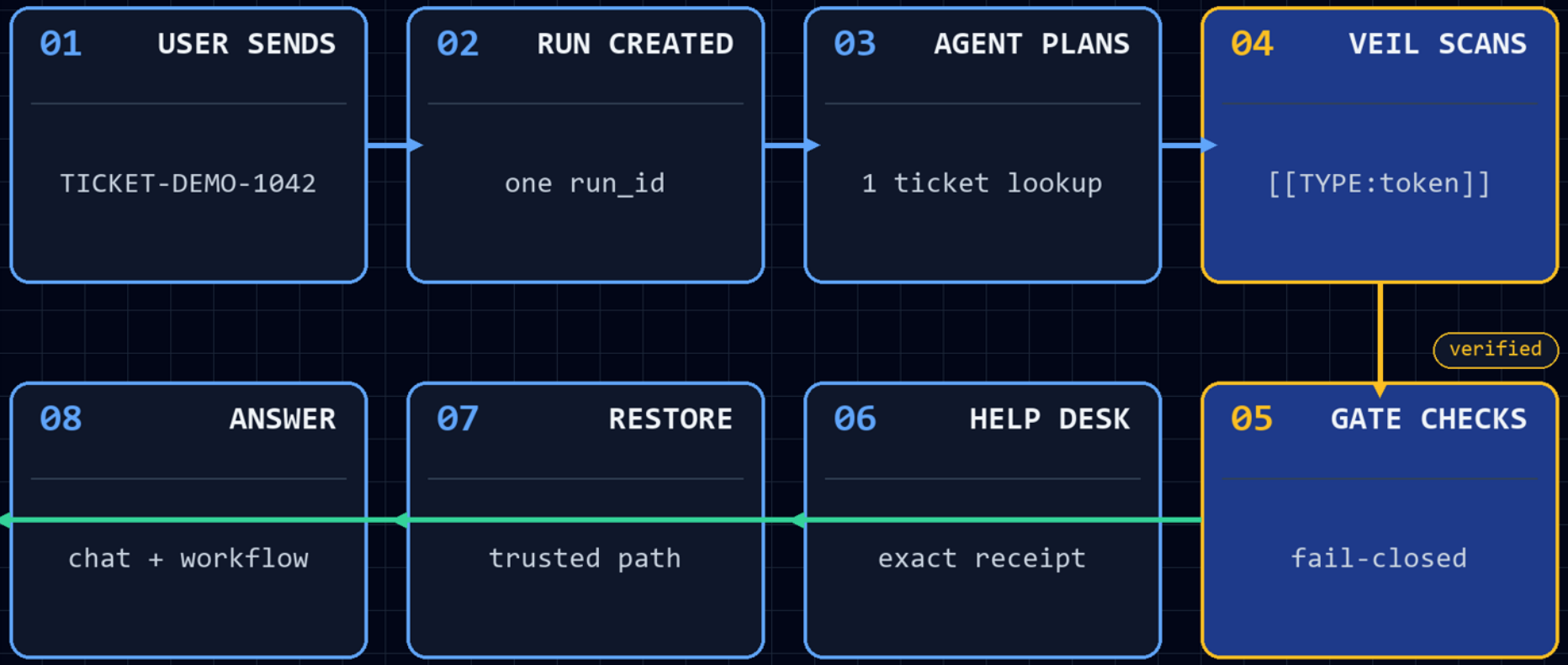
Tool answers in tokens. Vault maps back on the trusted path only.



Request lane left to right. Green return lane carries the tokenized reply home.

VEIL / ONE RUN, EIGHT STEPS

same run_id in chat + workflow // POST /api/runs once



TRIGGER send TICKET-DEMO-1042 + synthetic values

AHA open the proof - the receipt holds [[TOKENS]], not originals

CLOSE blocked example stops at the gate - 0 calls, nothing invented

Zero leaks.

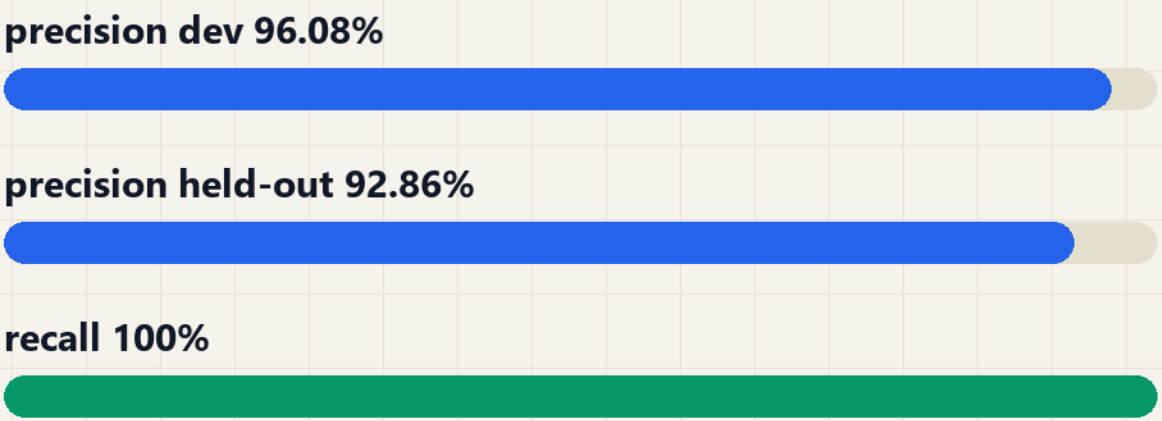
Measured, not claimed.

100%
recall dev + held-out

0
literal leaks at tool

99.2
F2 dev / 98.5 held-out

49+26
spans found



Dev 33 cases / 49 spans - held-out 26 / 26 - 2026-10-10. Cost admitted: 2 phone-lookalike FPs per split.

127.0.0.1:5173 /

Chat - valid ticket, real answer

[paste screenshot]

127.0.0.1:5173 /runs/{id}

Workflow - six stages, same run

[paste screenshot]

proof panel

What the helpdesk actually received

[paste screenshot]

blocked + streamlit

Gate stops it // nested-JSON reuse

[paste screenshot]

What it is not.

— Selected patterns only - the unknown stays unknown.

— Tokens are pseudonyms, not encryption.

— Names are the weakest recognizer.

— Synthetic data only. Logs keep counts, never values.



Two terminals. One ticket. Ninety seconds.

VEIL

Check the *receipt yourself.*

TICKET-DEMO-1042

deterministic // gemini optional // october 2026